



Multi-factor asset pricing models: Factor construction choices and the revisit of pricing factors

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ABSTRACT

We introduce the eight-factor asset pricing model as an extension of the Fama and French (2016b) five-factor model. In addition to capturing market premium, size, value, profitability and investment pricing factors, we propose three additional factors that represent momentum, liquidity and default risk. Albeit these factors are not new to the literature, our aim is to comprehensively and jointly test the performance of the model which accounts for all the suggested factors simultaneously. We find that the incorporation of additional factors improves the model's explanatory power. In addition to market, size and value factors, the profitability and momentum pricing factors exhibit higher explanatory power compared to investment, default risk, and liquidity pricing factors. The use of different stock allocation (number of sorting portfolios) and portfolio sorting approaches to factor construction has some, albeit statistically limited, effect.

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1. Introduction

The empirical evidence that CAPM model of Sharpe (1964) and Lintner (1965) leaves a lot of the variation in average returns unexplained has given a huge incentive to researchers to pursue the search for additional systematic risk factors that would better explain the variation in the cross-section of stock returns. The three-factor model of Fama and French (1993) is one of the key contributions in the development of multi-factor asset pricing models. In one of the most recent studies in this field, Fama and French (2016b) argue that in addition to the well-established three-factor model that captures size and value premium – next to the market risk premium – profitability and investment intensity should be considered as the additional two pricing factors. However, the model is still unable to account for a wide range of asset pricing anomalies. Thus, the search for an extended multi-factor model, which would be able to capture additional risk premia, is still ongoing.

Motivated by the extensive evidence of other “systematic anomalies”, we propose an extension of the most recent Fama-French five-factor model (Fama and French, 2016b) with three additional factors: momentum, liquidity risk and default risk factors. Momentum (Carhart, 1997) is one of the most prominent “anomalies” unaccounted for in either of Fama-French multi-factor models. Furthermore, the empirical research, particularly in the aftermath of the most recent global financial crisis, reveals the importance of liquidity, trading frictions and default risk in the cross-section of asset returns. Note, however, that through the remainder of the paper we generally consider risk factors and “anomalies” as pricing factors, thus staying away from the on-going debate whether the premia related to pricing factors should be seen

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as a risk-based phenomenon or is a consequence of pricing “anomalies” in the market. In contrast to some evidence that the firm bankruptcy is related to idiosyncratic risk (e.g. Opler and Titman, 1994; Asquith et al., 1994), we follow the argument that the default risk is systematic and, thus, should be considered as an additional pricing factor in asset pricing models (e.g. Vassalou and Xing, 2004; Garlappi and Yan, 2011; Friewald et al., 2014). In the construction of the default risk pricing factor, we use a simplified version of the structural model of credit risk as proposed by Bharath and Shumway (2008), the so-called naïve distance to default. We use the similar approach to examine the performance of the eight-factor model as in Fama and French (2016b). The analysis is performed on the sample of portfolio returns created from the universe of stocks contained in the Russell 3000 constituent list, which is a good representative of the US stock market.³ We use the same sample of portfolio returns to compare the performance of the five-factor model to the that of the proposed eight-factor model.

There are several contributions of the paper. The main contribution of this paper is the appraisal of the eight-factor asset pricing model, which, based on the latest empirical evidence, combines the pricing factors with the highest presumed explanatory power for average returns. The starting motivation for our research stems from the practical approach to corporate valuation. In practice, corporate appraisers and financial analysts typically use a standard CAPM model to determine the cost of equity, which is essentially a flip-side of what investors require in return compensation in order to invest in company's equity. This does not mean that other by now well established pricing factors do not affect corporate valuations in practice. What financial analysts typically do is to apply various discounts and premiums at the very end of the valuation process. The standard discounts are related to size and marketability, for example. Hence, the selected pricing factors most often appear as separate discounts to the value of the asset. From the perspective of financial theory, risks related to the asset (pricing factors) should be incorporated in the discount rate. The proposed eight-factor model thus enables the examination of the joint role of the selected pricing factors in explaining asset returns. Here, we also examine the performance of the extended versions of the Fama-French three-factor models with different factor combinations of our eight-factor model. In this way, we examine the relative impact of a particular pricing factor on the multi-factor model performance and the ability of five pricing factors incorporated in the Fama-French five-factor model (Fama and French, 2016b) to subsume momentum, liquidity and/or default pricing factor. Let us emphasize that our ambition here is not to extensively explore the “factor zoo”, as Cochrane (2011) referred the booming literature on pricing factors. Hence, the contribution of the paper should not be perceived from that perspective. Again, the first contribution of the paper is to use the existing evidence regarding the most persistent pricing factors (such as size, value and momentum) and extend it with additional two pricing factors for which we believe could have an important pricing impact in the cross-section of stock return. Our belief is, in particular, based on the corporate valuation practice of financial analysts.

While the main contribution is the examination of the joint role of pricing factors, which, based on the latest empirical evidence, should have the highest explanatory power of average returns, we are also interested in several data selection and factor model construction (estimation) issues. First, we investigate if and how choices in the factor model construction affect the results. More specifically, we address the impact of the portfolio sorting approach. The standard approach is to use the so-called independent sorting when creating portfolios of assets based on their characteristics (size, value, etc.). In such an approach, we are effectively looking at the effect of the both sorting variables on returns. In the dependent sorting approach, we are interested in the effect of the second sorting variable on returns, conditional on the first sorting variable. We believe that this issue might play an important role, since independent sorting could lead to very imbalanced portfolios from the perspective of number of assets (stocks) in each portfolio, even to an extent of making a particular portfolio non-investable in the most extreme case. Secondly, we investigate the effect of various breakpoints of sorting variables used to allocate stocks in portfolios. We expect that the choice of breakpoints, together with distributional properties of returns, to have an impact on the modelling outcome.

Based on the results of the GRS test (Gibbons et al., 1989) used to assess the performance of the proposed model, we argue that the three additional factors (i.e. momentum, liquidity and default risk factor) improve the model performance over the Fama-French five-factor model. The default risk pricing factor created from the naïve distance to default performs well in the eight-factor model, thus supporting our view that the default risk bears, in part, systemic characteristics as well. It seems that average returns on investment, momentum and liquidity factors are to a great extent captured by their exposures to other pricing factors. We find that the estimation choices in the construction of explanatory (RHS) portfolios have an impact on returns of some factors. Furthermore, the estimation choices in the creation of the testing (LHS) portfolios provide mixed conclusions about the overall performance of the multi-factor models, but they do affect the explanatory power of the pricing factors in the regression model.

The paper is structured as follows. We begin with a review of empirical evidence regarding the pricing factors and the presentation of multi-factor models inspired by these factors in Section 2. In Section 3 we present the building blocks of the proposed eight-factor model and describe the data. In Section 4 we start our analysis with the portfolio sorting approach in order to investigate return patterns. We then continue with the investigation of how the pricing factor construction

³ It consists of 3000 largest companies in the U.S. equity market, representing approximately 98% of the investable U.S. equity market. For comparison, the CRSP US Total Market Index covers 3700 companies, including nearly all publicly traded U.S. stocks. Hence, the Russell 3000 index constituent list makes for a very good representation of the investable equities in the U.S. equity market. Moreover, one could argue that the remaining part of public equity market not covered by the Russell 3000 index consists of the smallest public equities, which (1) are relatively unimportant from the aggregate perspective and (2) might be very difficult to invest in.

choices affect the factor estimates in Section 5. In Section 6 we deal with the estimation of factor loadings that represent the input for further analysis. The regression results wrap up the analysis. Finally, we summarize the main findings in Section 7 and conclude.

2. Previous research

2.1. Empirical evidence of pricing factors

This study relates to the extensive empirical literature on asset pricing models. In line with the risk and return paradigm, the asset pricing models aim to explain the variation in asset returns with a set of factors, which represent the compensation for non-diversifiable risks. In the seminal study in this field, [Sharpe \(1964\)](#) and [Lintner \(1965\)](#) present the capital asset pricing model (CAPM), where a single (market) factor is used to explain the cross-section of average asset returns. The subsequent empirical studies show that the market factor does not capture all the relevant risks associated with asset returns (see for example [Banz, 1981](#); [Fama and French, 1992](#); [Fama and French, 2004](#)). The extensive search for additional risk factors, which would better explain the cross-section of average asset returns, has resulted in a wide range of different pricing factors that appear in multi-factor models ([Jagannathan et al., 2010](#); [Subrahmanyam, 2010](#); [Cochrane, 2011](#)). In this paper, we mostly focus on pricing factors based on firms' fundamentals, as these are the most commonly used in valuation models in practice.

Since the late 1970s many studies have documented the so-called “anomalies” related to the CAPM model. The empirical tests have shown that various firm characteristics and market frictions can (additionally) explain the variation in average returns. [Basu \(1977\)](#) and [Ball \(1978\)](#) show that firms with low price-earnings ratio on average earn higher returns than firms with high price-earnings ratio. [Banz \(1981\)](#) documents the firm size effect on average returns by showing that returns on small stocks, measured by market capitalization, are higher than returns on big stocks. The research of [Stattman \(1980\)](#) and [Rosenberg et al. \(1985\)](#) show that firms with high book-to-market (B/M) value of equity ratio earn higher returns. [Chan et al. \(1991\)](#) present the same conclusion for the sample of Japanese stocks. Similar evidence about the relationship between firm capital structure and average returns is provided by [Bhandari \(1988\)](#), who shows that firms with higher leverage, measured as the ratio between the book value of debt and the market value of equity, earn higher returns than firms with low leverage. In one of the most influential (and to-date cited) studies, [Fama and French \(1992\)](#) summarize the findings from earlier studies regarding the impact of the market factor, the size effect, the earnings-price (E/P) ratio and the B/M ratio (valuation effect), as well as the leverage (financial risk effect) in the cross-section of average stock returns. They show that the size and the B/M ratio seem to absorb the impact of the E/P ratio and the leverage on average returns.

In addition, a broad literature examines “anomalies” in terms of departure of observable returns from the ones predicted by the CAPM model, this posing a significant challenge to the CAPM and the efficient markets hypothesis (EMH). These studies provide the evidence of momentum effect in average returns (e.g. [Jegadeesh and Titman, 1993](#); [Jegadeesh and Titman, 2001](#); [Chordia and Shivakumar, 2002](#); [Griffin et al., 2003](#); [Moskowitz et al., 2012](#)), as well as the long-term reversal effect ([De Bondt and Thaler, 1985](#); [De Bondt and Thaler, 1987](#)). Many authors attempt to explain the “anomalies” in the cross-section of returns with accounting variables. [Lakonishok et al. \(1994\)](#), [Porta and R. \(1996\)](#), [Haugen and Baker \(1996\)](#), [Cohen et al. \(2002\)](#) and others examine the relation between the average returns and financial performance of firms measured by profitability and sales growth. Another group of studies presents the impact of investment intensity, however showing very mixed evidence about the relationship ([Chan et al., 2001](#); [Cooper et al., 2008](#)).

Most of the asset pricing models are based on frictionless markets. However, given the actual characteristics of the markets and the market microstructure, many authors investigate how various measures of liquidity, as well as the relationship between stock liquidity and market liquidity affect average returns ([Chordia et al., 2000](#); [Pastor and Stambaugh, 2003](#); [Acharya and Pedersen, 2005](#); [Brunnermeier and Pedersen, 2009](#)).

2.2. Multi-factor models inspired by pricing factors

Based on the ample evidence regarding the strength and importance of various pricing factors, [Fama and French \(1993\)](#) propose a three-factor model with size (SMB) and value (HML) factors (in addition to the market factor) to explain the cross-section of returns. They show that the three factors are able to explain most of the variation in double sorted size and B/M portfolio returns. However, the model is still unable to account for all the determinants of returns and leaves a significant portion of the variation in returns unexplained, particularly in small growth portfolios. The subsequent studies mainly attempt to improve the explanatory power of the three-factor model by introducing additional factors as well as attempting to explain which risks are represented by size and B/M factors.

In addition, a wide a range of subsequent studies document that Fama-French three-factor model seems to be unable to account for a wide range of anomalies (e.g. [Jegadeesh and Titman, 1993](#); [Ang et al., 2006](#); [Daniel and Titman, 2006](#); [Campbell et al., 2008](#)). Inspired by the evidence of momentum effect on stock returns ([Jegadeesh and Titman, 1993](#)), [Carhart \(1997\)](#) explains the mutual fund performance patterns with a four-factor model, which extends the Fama-French three-factor model with a factor representing one-year momentum in stock returns. [Pastor and Stambaugh \(2003\)](#) investigate whether

the average returns are related to systematic liquidity risk in returns. They model the impact of liquidity risk on average returns by extending Fama-French three-factor model with a liquidity factor.

Inspired by the investment-based asset pricing principles and the available evidence regarding the anomalies of Fama-French three-factor model, [Hou et al. \(2015\)](#) construct a new empirical model that summarizes the cross-section of average stock returns and captures many of the pricing factors that remain unaccounted by Fama-French three-factor and [Carhart \(1997\)](#) four-factor models. In their model, the excess returns on an asset are described by the sensitivities to market factor, size factor, investment factor and profitability factor, which is measured by return on equity (ROE).

Many studies demonstrating that Fama-French three-factor model's inability to explain much of the variation related to profitability and investment intensity ([Haugen and Baker, 1996](#); [Cohen et al., 2002](#); [Titman et al., 2004](#); [Fama and French, 2008](#); [Novy-Marx, 2013](#)) motivated [Fama and French \(2016b\)](#) to revisit their three-factor model by extending it with two additional factors, namely investment and profitability factors. The model performs better than the three-factor model, although it fails to explain the average returns on small stocks with the same return patterns, such as low profitable firms with high investment intensity, for example.

3. The model and the data

As mentioned, [Fama and French \(2016a\)](#) extended the three-factor model with profitability and investment factors:

$$R_{it} - R_{ft} = \alpha_i + \beta_{1,i}(R_{mt} - R_{ft}) + \beta_{2,i}SMB_t + \beta_{3,i}HML_t + \beta_{4,i}RMV_t + \beta_{5,i}CMA_t + \epsilon_{it} \quad (1)$$

In Eq. (1) R_{it} represents the return on security i in period t , R_{mt} is the market return, R_{ft} is the risk free rate, SMB_t is the return on a diversified portfolio of small minus big stocks, HML_t is the difference between returns on diversified portfolios of high and low book-to-market stocks, RMV_t is the difference between the returns on diversified portfolios of conservative and aggressive firms from the investment perspective, while CMA_t represents the difference in returns on portfolios of stocks with robust and weak profitability and the ϵ_{it} is the residual. The parameters $\beta_{2,i} - \beta_{5,i}$ represent the sensitivity of returns to the five factors respectively.

Based on the empirical evidence that momentum effect ([Carhart, 1997](#)), liquidity risk ([Pastor and Stambaugh, 2003](#)) and default risk ([Vassalou and Xing, 2004](#)) affect the variation in average equity returns, we extend the [Fama and French \(2016b\)](#) five-factor model with additional three factors representing momentum, liquidity and default risk pricing factors:

$$R_{it} - R_{ft} = \alpha_i + \beta_{1,i}(R_{mt} - R_{ft}) + \beta_{2,i}SMB_t + \beta_{3,i}HML_t + \beta_{4,i}RMV_t + \beta_{5,i}CMA_t + \beta_{6,i}WML_t + \beta_{7,i}IML_t + \beta_{8,i}BMS_t + \epsilon_{it} \quad (2)$$

In Eq. (2) WML_t represents the momentum effect as the difference in returns on portfolios of winners and losers, IML_t captures the difference in returns on portfolios of illiquid and liquid stocks, whereas BMS_t represents the difference in average return on portfolios of stocks with high and low default risk, and the ϵ_{it} is the residual. The parameters $\beta_{6,i} - \beta_{8,i}$ represent the sensitivity of returns to the additional three factors respectively.

We introduce BMS factor into our model to incorporate the default risk in explanation of equity returns. [Vassalou and Xing \(2004\)](#) examine whether default risk is priced in equity returns with the probability implied by the Merton's distance to default (DD) model, which is based on Merton's ([Merton, 1974](#)) corporate debt (structural) pricing model. The Merton's DD model has a firm theoretical basis, but it requires many strong assumptions, such as geometric Brownian motion of the underlying firm's value and a zero coupon bond issued by the firm, among others. The restrictive assumptions of the model and relatively high complexity to calculate distance to default were the motivation to use the naïve version of the Merton's model proposed by [Bharath and Shumway \(2008\)](#). The naïve alternative is easier to calculate and it also captures the functional form of Merton DD model. Thus, the distance to default calculated with the naïve alternative should adequately represent the default risk pricing factor in the proposed multi-factor asset pricing model.

The empirical analysis examines whether the proposed eight-factor model better explains the average returns on portfolios of stocks than multi-factor models with different subsets of factors of the eight-factor model. In the construction of portfolios we use 3,000 US stocks included in the Russell 3000 Index constituent list, which represent approximately 98% of all investable US equity market. We retrieve the data from the Datastream online database of Thomson Reuters. The data spans the period between January 1985 and December 2016. The Russell 3000 price index values in [Fig. 1](#) show the general price movement of sample stocks in this period. The constant growth of stock prices between 1985 and 1995 continued with a period of significant stock price volatility between 1995 and 2016. Two significant negative trends in average returns in the periods from September 2000 until September 2002 and from November 2007 until February 2009 interrupted quite intensive average growth of stock prices between 1995 and 2016. In comparison to some other related studies, the shorter time period of sample data in this study puts somewhat more emphasis on recent market trends. In addition, while the sample period might be shorter than in a comparable studies, it still spans the period of more than 30 years. Having said that, we nevertheless had some issues when it comes to the data availability in the initial years of our sample period, thus limiting the actual data availability and the sample size when investigating all eight pricing factors.



Fig. 1. Russell 3000 – Price Index, Jan 1985–Dec 2016.

4. Pricing factors and return patterns

We test the eight-factor model and multi-factor models constructed with different subsets from eight factors on six sets of 25 portfolios formed by sorts on size and the second sorting variables. The first step is to analyze the return patterns for each set of 25 portfolios characterized by the two selected sorting variables. We follow a similar approach to the one used by Fama and French in their studies of three-factor and five-factor model (Fama and French, 1993; Fama and French, 2016b). At the end of each year the stocks are allocated independently into five size, value, profitability, investment, momentum, liquidity and default risk groups. The fundamental variables, which characterize the first four groups, are stock market capitalization, book-to-market (B/M) ratio, operating profitability and annual percentage growth of total assets, respectively. They correspond to the variables used by Fama and French (2016b), except for the variable operating profitability. In our study it is represented by the EBIT margin (i.e. EBIT-to-sales ratio), whereas Fama and French (2016b) calculate this variable as sales revenues minus cost of goods sold, interest expense and selling, general and administrative expenses, all divided by book value of equity. We use the stock return in the most recent 12 months, the ratio between the average number of shares traded in particular month and total shares of the company, and the naïve distance to default to determine momentum, liquidity and default risk groups respectively.

Table 1 shows the average monthly percent returns in excess of the three months US Treasury bill rate for six sets of 25 portfolios formed from the independent sorts of size and the other six variables. Panel A of Table 1 shows the average monthly excess returns for 25 portfolios based on five size and five B/M groups (i.e. 5×5 Size-B/M portfolios). The results support the empirical predictions of returns increasing from small to big stocks, the so-called size effect, for all B/M groups, albeit not with monotonic increases for all the B/M groups. The return spread between the small and big portfolios is the highest for the value portfolios, i.e. portfolios with the highest value of B/M ratio, and it decreases along portfolios with the falling B/M ratio. Moreover, value portfolios show strong presence of the size effect across all size groups, whereas the relation between the size and the value is not so obvious for the growth stocks (low B/M).

The average excess returns on 25 portfolios sorted on size and B/M in Panel A of Table 1 show more consistent presence of the value effect. Namely, the returns increase from portfolios with the low B/M stocks to portfolios with high B/M stocks across all five size groups. In line with the observed return patterns by Fama and French (2016b), the value effect is stronger among small stocks. The portfolio of small stocks with the lowest B/M earned a 0.67% average monthly excess return whereas the average return on the portfolio with big stocks and the highest B/M amounted to 1.18%. In the big stock group the average monthly excess returns increase from 0.41% in the lowest B/M portfolio to 0.70% in the highest B/M portfolio.

Panel B of Table 1 shows the average monthly excess returns for 25 portfolios formed on the intersections between size and profitability quintiles (i.e. 5×5 Size-Profitability portfolios). The size effect is present in all five profitability groups. However, the decline in average monthly excess returns from small to big stocks is not monotonic across all size groups in each profitability column. The difference between returns on small and big stock portfolios is the highest for the average profitability portfolios ($1.33 - 0.42 = 0.91\%$ and $1.06 - 0.39 = 0.67\%$). The examination of the profitability effect reveals similar patterns in average returns as in some previous empirical studies (Novy-Marx, 2013; Fama and French, 2016b). Namely, the average excess returns typically increase with higher profitability of the firm. The average excess monthly returns are almost monotonically increasing across all profitability portfolios in the size groups of bigger stocks. Despite the significant difference in average excess returns between the portfolios with high and low profitability, the relation between average excess returns and profitability is not obvious across profitability portfolios in the small stock groups.

Table 1

Average monthly percent excess returns for 5×5 portfolios formed on independent sorts of Size and Book-to-market ratio; Size and Profitability ratio; Size and Investment ratio; Size and Momentum ratio; Size and Liquidity ratio; Size and Default risk (distance to default ratio) in the period January 1985 – December 2016.

	Low	2	3	4	High
<i>Panel A: Size - B/M portfolios</i>					
Small	0.67	0.45	0.66	0.71	1.18
2	0.13	0.19	0.43	0.67	0.74
3	0.08	0.37	0.51	0.58	0.67
4	0.28	0.43	0.53	0.63	0.70
Big	0.41	0.39	0.48	0.35	0.70
<i>Panel B: Size - Profitability portfolios</i>					
Small	0.56	1.33	1.06	0.82	0.93
2	0.14	0.52	0.63	0.52	0.70
3	0.00	0.48	0.54	0.55	0.58
4	-0.07	0.46	0.66	0.68	0.45
Big	0.30	0.42	0.39	0.44	0.55
<i>Panel C: Size - Investment portfolios</i>					
Small	0.97	0.71	0.90	0.90	0.80
2	0.43	0.57	0.59	0.60	0.26
3	0.44	0.59	0.58	0.45	0.18
4	0.44	0.57	0.57	0.53	0.28
Big	0.51	0.57	0.61	0.28	0.13
<i>Panel D: Size - Momentum portfolios</i>					
Small	0.57	0.51	0.71	1.08	1.69
2	-0.02	0.46	0.72	0.62	0.76
3	-0.05	0.50	0.50	0.61	0.57
4	-0.01	0.54	0.72	0.58	0.39
Big	-0.04	0.52	0.63	0.51	0.31
<i>Panel E: Size - Liquidity portfolios (volume turnover)</i>					
Small	0.91	1.08	0.94	0.85	0.24
2	0.51	0.71	0.62	0.40	0.08
3	0.58	0.66	0.59	0.38	0.12
4	0.55	0.58	0.71	0.42	0.28
Big	0.34	0.47	0.52	0.52	0.20
<i>Panel F: Size - Default risk portfolios</i>					
Small	0.71	0.84	0.84	1.24	0.94
2	0.20	0.46	0.58	0.63	0.64
3	-0.03	0.41	0.60	0.59	0.60
4	0.07	0.47	0.46	0.63	0.63
Big	0.05	0.45	0.52	0.46	0.48

Panel C of [Table 1](#) shows the average monthly excess returns for 25 portfolios based on the size and investment factor quintiles (i.e. 5×5 Size-Investment portfolios). The size effect is more evident for portfolios of firms with the highest investment intensity (quintiles 4 and 5). The five portfolios in the smallest size quintile earned much higher returns than portfolios in other size quintiles. However, the relation between average returns and the firm size is less evident for the other four size groups. The overview of return patterns by investment quintiles shows an important impact of the investment intensity on average monthly excess returns. Firms with low investment intensity on average earned higher returns in each size group. The difference in returns between firms with high and low investment intensity is more pronounced for the small stock portfolios.

In Panels D, E, and F we present the extension of the [Fama and French \(2016b\)](#) average portfolio return analysis by replacing the second sorting variable in the creation of 25 portfolios with either momentum, liquidity or distance-to-default pricing factor. In Panel D of [Table 1](#) we show the average monthly excess returns for 25 size and momentum portfolios. The average portfolio returns support the size effect in all momentum quintiles. However, the relation between average returns and size is much stronger for portfolios of winners. The average monthly returns on portfolios of the best performing stocks in the last 12 months consistently decline from 1.69% for the small stock portfolio to 0.31% for the big stock portfolio. The analysis of portfolio returns in each size quintile reveals higher average monthly returns for portfolios of stocks with higher average past 12-month returns (i.e. momentum effect). Holding size constant, the momentum effect is more evident in the small stock than in the big stock quintiles. In latter groups the average returns are higher for the mid quintiles momentum groups.

In Panel E of [Table 1](#) we report the average monthly excess returns for 25 portfolios formed based on five size and five liquidity quintiles (i.e. 5×5 Size-Liquidity portfolios). The size effect shows up in all liquidity quintiles except for the portfolios consisting of the most liquid stocks. The difference between the portfolio returns of small and big stocks is the highest in the two low momentum quintiles. The results reveal the impact of stock liquidity on average returns. Holding size roughly

constant, the average returns typically fall as liquidity increases. The liquidity effect is much stronger in the small stock portfolio quintiles. Namely, the average returns in the smallest stock quintile fall from 0.91% for illiquid stocks to 0.24% for liquid stocks (a decrease of 0.67%), whereas the difference between illiquid and liquid portfolios in the biggest stock portfolio quintile amounts to only 0.14% with less obvious decline across all portfolios in this size quintile.

Panel F of Table 1 we show the average monthly excess returns for 25 portfolios based on the independent sorts of stocks into size and default risk portfolios (i.e. 5×5 Size-Default risk portfolios). The average return patterns confirm the presence of the size effect in all the default risk quintiles. The declining returns with increasing firm size are more consistent for the portfolios of stocks with higher default risk. The analysis of the relation between the average returns and the default risk reveals patterns in contrast to our expectations. In every size quintile, the average returns typically increase with increasing distance-to-default and thus decreasing default risk. However, the strength of the »default risk effect« is roughly similar in all size quintiles. The difference in returns between the lowest and the highest default risk portfolios amounts to 0.23% for the smallest stock quintile and 0.44% for the biggest stock quintile.

The selection of a particular single variable next to size in the creation of the six portfolio sets used to examine the impact of firm variables on average portfolio returns does not exclude an impact of other omitted variables on average returns. Fama and French (1995, 2016a) show that B/M ratio, operating profitability and investment pricing factors are correlated. For example, they argue that low B/M growth firms tend to be profitable and have high investment intensity. The introduction of three additional variables in the analysis would increase the complexity of measuring the interrelations among all seven variables. The multiple sorts with seven variables would result in large number of portfolios consisting of low number of stocks (in some cases even empty portfolios) and consequently lower diversification. Hence, we need to revert to a different approach in exploring the cross-correlation between the pricing factors, which we will revisit later.

5. Pricing factors' construction choices

The relationships between average monthly excess returns and the selected firm variables represents the foundation for the creation of pricing factors - the explanatory or the right hand side (RHS) variables in the eight-factor model. As mentioned earlier, we follow a similar approach to factor creation as in Fama and French (1993, 2016b). Each month we calculate factors from monthly returns on portfolios of stocks specifically created for the factor construction. We create these portfolios by using two approaches to allocate stocks into different portfolios. According to the first approach, we allocate stocks into two size groups and three groups of the second sorting variable (i.e. 2×3 sorts), whereas under the second approach we split stocks into two size and two second variable groups (i.e. 2×2 sorts). The option to perform the second variable sorts independently or dependently given the sorting based on the first variable finally results in four different versions of a particular pricing factor.

In the standard (independent sorting) approach we are implicitly investigating the effect of the both sorting variables on average returns. On the other hand, with the dependent sorting we are exploring the effect the second sorting variable has for the portfolio returns, conditional on the sorting according to the first sorting variable. The two approaches clearly lead to different sorting outcomes, even more so since the dependent sorting approach is not sort-order invariant. Moreover, independent sorting could lead to very imbalanced portfolios from the perspective of number of assets (stocks) in each portfolio, potentially making a particular portfolio even non-investable. Secondly, we explore how various number of breakpoints for sorting variables (one or two), which is ultimately an arbitrary decision, affects the factor construction. We wish to emphasize that it is not our aim to propose the best of the various options, but simply to explore if and to what extent do the described choices in the pricing factors' construction affect the findings (results).

Market factor is the first factor in the eight-factor model that represents the average monthly excess return of all the stocks in the sample (over the 3-months US Treasury bill rate). The estimation choices in the creation of portfolios do not have any impact on the value of this factor. The creation of other seven factors starts with the proposed allocation of stocks into different groups. At the end of each year we allocate stocks into two size and three groups for each of the remaining six sorting variables. The breakpoint to split stocks into two size groups is the median market capitalization, whereas the 30th and 70th percentiles represent the breakpoints for the other six variables. The stocks are then re-sorted again at the end of next year. The sorting based on the two size groups and three B/M groups yields six portfolios (i.e. independent 2×3 portfolios). Replacing the B/M group with the profitability, investment, momentum, liquidity or default risk groups results in total six sets of independent 2×3 portfolios.

The size factor *SMB* represents the average monthly return of portfolios of small minus big stocks. We calculate *SMB* factor from the double sorted portfolios for each of the six sets of 2×3 portfolios as the average return of the three small stock portfolios minus the average return on the three big stock portfolios. In this way, we end up with six different *SMB* factors in a particular month (i.e. $SMB_{Size} - SMB_{Default}$). The *SMB* factor used in further analysis represents the average of these six *SMB* factors.

The value factor *HML* represents the average return of the two high B/M portfolios minus the average return of the two low B/M portfolios in 2×3 sorts on Size and B/M. Similarly as before, the value factor can be defined as the average of the two value factors calculated separately only on portfolios of small stocks (*HML_{Small}*) or only on big stocks (*HML_{Big}*). We apply the same principle to calculate the remaining factors of the eight-factor model. The profitability factor *RMW* represents the average return of portfolios of stocks with a robust (high) profitability minus the average return of portfolios with a weak

profitability. The investment factor *CMA* is based on the average return of portfolios with a conservative investment policy minus the average return of portfolios with an aggressive investment policy, both determined by the annual percentage growth of total assets. The momentum factor *WML* is constructed as the average return of portfolios of winners minus average return of portfolio of losers based on the most recent 12-month stock return. The liquidity factor *IML* represents the difference in average returns of illiquid and liquid stock portfolios. The default factor *BMS* is the difference in the average return of bankruptcy portfolios, which consist of stocks with small distance-to-default, and the average returns of safe portfolios, which are represented by companies with the large distance-to-default.

The definition of factors from 2×2 portfolio sorts follows the same principles as applied in the factor construction based on 2×3 portfolio sorts. The only difference is the allocation of stocks into two groups for the second sorting variable, where now the median value of the sorting variable in the cross-section is used as a sorting breakpoint. We repeat the same procedure and create additional set of 2×3 and 2×2 portfolios based on the dependent sorting approach.

In Table 2 we present the summary statistics for factor returns. In Panel A we report factor returns based on the independent 2×3 and 2×2 sorts, whereas in Panel B factor returns from the dependent 2×3 and 2×2 sorts are presented. In Panel C we show correlations between different versions of the same factor and in Panel D we report correlations among different factors of the eight-factor model.

The average monthly market return in excess of the three month US Treasury bill ($R_m - R_f$) amounts to 0.94% with the standard deviation of 4.44% in the sample period between January 1985 and December 2016. The calculation of the market factor is not related to the sorting of stocks in portfolios, thus the results are the same for all the implemented approaches to factor construction (Panels A and B of Table 2). The average *SMB* factors' return amounts to 0.15% and 0.16% for 2×3 and 2×2 portfolios, respectively. The construction of the size factor *SMB* considers all sample stocks in their allocation to portfolios, which explains the small difference in the reported factor values. The relatively low correlation of the size factor with other explanatory factors also contributes to small variation of the reported results for dependent and independent sorts. The standard deviations of the *SMB* factors for the independent 2×3 and 2×2 portfolios are 1.79% and 1.74%, whereas the standard deviations for dependent 2×3 and 2×2 portfolios amount to 1.88% and 1.80%, respectively. The results for the *SMB* factor support the presence of size effect for sample stocks, however, the observed t-statistics between 1.55 and 1.77 reveal lower statistical significance than observed in previous reference studies (Fama and French, 1993, 2016b).

The results for *HML*, *RMW*, *CMA*, *WML*, *IML* and *BMS* factors depend on the estimation choices used in the calculation of factors. In general, the average factor returns are higher for the 2×3 stock portfolios than 2×2 portfolios. We believe that this is due to fact that the first approach to stock allocation considers only more extreme values of the second sorting variable by excluding the middle 40% of stocks between 30th and 70th percentile. However, the downside of excluding some stocks in portfolio allocation is lower portfolio diversification and thus on average higher standard deviation of factors based on the 2×3 sorting portfolios compared to 2×2 sorts.

The results show that the largest effect related to the choice of breakpoints can be observed in the case of momentum and profitability factors. The independent sorting of stocks yields a difference in calculated *WML* and *RMW* factors based on 2×3 and 2×2 portfolios of 0.18 and 0.11 percentage points respectively, whereas the differences in these two factors calculated by applying dependent sorting amounts to 0.18 and 0.16 percentage points, respectively. The choice of independent versus dependent sorting of stocks has the largest impact on profitability (*RMW*) and liquidity factors (*IML*) in the case of 2×3 stock portfolios. In the case of 2×2 stock portfolios, the choice between independent and dependent sorting has an important effect in the case of the value factor (*HML*), in addition to the already mentioned profitability and liquidity factors.

Despite some impact of the implemented stock allocation (number of sorting portfolios) and portfolio sorting approaches on factor returns, the average monthly factor returns and t-statistics in Table 2 provide similar conclusions regarding the size and statistical significance of factor spreads. The average monthly market premium amounts to 0.94% ($t = 4.14$). We observe the largest spreads for the momentum factor, ranging between 0.30% ($t = 2.35$) and 0.48% ($t = 2.79$). The average monthly value factor spreads (*HML*) are between 0.22% ($t = 3.25$) and 0.33% ($t = 3.76$), while the average monthly profitability factor spreads (*RMW*) range from 0.13% ($t = 1.43$) to 0.32% ($t = 2.60$). In terms of the size factor we observe the average monthly spreads are from 0.15% ($t = 1.55$) to 0.16% ($t = 1.77$), whereas the average monthly liquidity factor spreads (*IML*) amount to between 0.15% ($t = 1.10$) and 0.30% ($t = 1.70$) - in both cases statistically insignificant. The summary statistics for factor returns show investment factor (*CMA*) returns between 0.06% and 0.09% with t-statistics lower than 1. The average monthly default risk factor premia (*BMS*) are between -0.18% ($t = -1.81$) and -0.29% ($t = -2.15$).

The calculation of *HML*, *RMW*, *CMA*, *WML*, *IML* and *BMS* factors on subsamples of only small or big stocks corroborate the conclusions regarding the relationship between size and the second sorting variable from the earlier 5×5 portfolio sorting analysis. The absolute average factor returns are larger for subsamples of small stocks, except for the investment factor (*CMA*). Table 2 shows the largest difference in average factor returns between small and big stock subsamples for the value factor (*HML*). For independent portfolio sorting approach the value factors *HML_S* are 0.61% ($t = 4.12$) and 0.40% ($t = 3.93$) for 2×3 and 2×2 stock portfolios respectively, whereas the value factors *HML_B* are only in the range between 0.30% ($t = 2.66$) and 0.57% ($t = 2.96$) for the same two stock allocation groups. The dependent portfolio sorting approach produces similar differences in average factor returns on small and big stock portfolios, which confirms the earlier evidence of larger value premium for small stocks (Fama and French, 1993; Fama and French, 2016b). Furthermore, the comparison of factor returns of small and big stock portfolios indicates stronger momentum effect for small stock portfolios. The average differences between *WML_S* and *WML_B* are 0.24% ($t = 1.83$) and 0.28% ($t = 1.69$) for two independent stock sorted groups. The dependent

Table 2

Summary statistics for monthly factor percent returns, January 1985 – December 2016 (N = 384).

Panel A: Averages, standard deviations and t-statistics for monthly returns - Independent sorts													
	$R_m - R_f$	SMB	HML	RMW	CMA	WML	IML	BMS					
2 × 3 Factors													
Mean	0.94	0.15	0.33	0.24	0.09	0.48	0.30	−0.27					
Std dev.	4.44	1.79	1.75	2.54	2.08	3.55	3.45	2.81					
t-statistics	4.14	1.67	3.68	1.83	0.85	2.65	1.70	−1.90					
2 × 2 Factors													
Mean	0.94	0.16	0.22	0.13	0.08	0.30	0.21	−0.18					
Std dev.	4.44	1.74	1.30	1.73	1.62	2.52	2.67	1.94					
t-Statistics	4.14	1.76	3.25	1.43	0.93	2.35	1.53	−1.81					
	HML_L	HML_H	HML_{L-H}		RMW_S	RMW_B	RMW_{S-B}		CMA_S	CMA_B	CMA_{S-B}		
2 × 3 Factors													
Mean	0.61	0.05	0.57		0.35	0.13	0.22		0.02	0.16	−0.14		
Std dev.	2.91	2.16	3.74		3.76	2.32	3.63		2.47	2.69	3.05		
t-Statistics	4.12	0.43	2.96		1.81	1.07	1.19		0.18	1.16	−0.87		
2 × 2 Factors													
Mean	0.40	0.03	0.37		0.21	0.04	0.18		−0.02	0.18	−0.20		
Std dev.	2.01	1.79	2.78		2.72	1.70	2.95		1.79	2.24	2.46		
t-Statistics	3.93	0.31	2.64		1.54	0.43	1.18		−0.26	1.55	−1.60		
	WML_S	WML_B	WML_{S-B}		IML_S	IML_B	IML_{S-B}		BMS_S	BMS_B	BMS_{S-B}		
2 × 3 Factors													
Mean	0.62	0.34	0.28		0.37	0.23	0.14		−0.29	−0.25	−0.04		
Std dev.	4.18	3.62	3.28		4.15	3.25	2.79		3.33	3.06	3.06		
t-Statistics	2.91	1.83	1.69		1.74	1.40	0.96		−1.73	−1.61	−0.28		
2 × 2 Factors													
Mean	0.42	0.18	0.24		0.22	0.19	0.03		−0.26	−0.10	−0.16		
Std dev.	2.96	2.68	2.54		3.10	2.64	2.17		2.40	2.19	2.45		
t-Statistics	2.79	1.34	1.83		1.42	1.42	0.30		−2.10	−0.90	−1.26		
Panel B: Averages, standard deviations and t-statistics for monthly returns - Dependent sorts													
	$R_m - R_f$	SMB	HML	RMW	CMA	WML	IML	BMS					
2 × 3 Factors													
Mean	0.94	0.15	0.32	0.32	0.08	0.48	0.20	−0.29					
Std dev.	4.44	1.88	1.69	2.39	2.11	3.37	3.53	2.65					
t-statistics	4.14	1.55	3.73	2.60	0.72	2.79	1.10	−2.15					
2 × 2 Factors													
Mean	0.94	0.16	0.33	0.16	0.06	0.30	0.15	−0.20					
Std dev.	4.44	1.80	1.72	1.65	1.66	2.43	2.71	1.88					
t-Statistics	4.14	1.77	3.76	1.92	0.68	2.43	1.11	−2.13					
	HML_L	HML_H	HML_{L-H}		RMW_S	RMW_B	RMW_{S-B}		CMA_S	CMA_B	CMA_{S-B}		
2 × 3 Factors													
Mean	0.56	0.08	0.48		0.55	0.08	0.47		0.01	0.14	−0.13		
Std dev.	2.44	1.96	2.85		3.93	2.12	4.11		2.55	2.70	3.14		
t-Statistics	4.54	0.80	3.33		2.77	0.74	2.26		0.09	1.03	−0.81		
2 × 2 Factors													
Mean	0.33	0.03	0.30		0.30	0.02	0.28		−0.04	0.16	−0.20		
Std dev.	1.72	1.61	2.18		2.71	1.72	3.11		1.81	2.29	2.45		
t-Statistics	3.76	0.39	2.66		2.20	0.22	1.79		−0.47	1.36	−1.62		
	WML_S	WML_B	WML_{S-B}		IML_S	IML_B	IML_{S-B}		BMS_S	BMS_B	BMS_{S-B}		
2 × 3 Factors													
Mean	0.72	0.24	0.47		0.16	0.23	−0.07		−0.43	−0.15	−0.28		
Std dev.	4.15	3.33	3.35		4.23	3.33	2.84		3.50	2.73	3.34		
t-Statistics	3.38	1.44	2.76		0.76	1.37	−0.47		−2.41	−1.09	−1.63		
2 × 2 Factors													
Mean	0.46	0.14	0.32		0.09	0.22	−0.13		−0.33	−0.08	−0.24		
Std dev.	2.88	2.63	2.60		3.02	2.79	2.11		2.46	2.07	2.55		
t-Statistics	3.14	1.05	2.41		0.58	1.53	−1.20		−2.60	−0.79	−1.87		
Panel C: Correlation coefficients among different versions of pricing factors													
		HML				RMW				CMA			
		Independent		Dependent		Independent		Dependent		Independent		Dependent	
		2 × 3	2 × 2	2 × 3	2 × 2	2 × 3	2 × 2	2 × 3	2 × 2	2 × 3	2 × 2	2 × 3	2 × 2
Independent	2 × 3	1.00	0.95	0.93	0.73	1.00	0.96	0.98	0.95	1.00	0.97	0.99	0.96
	2 × 2	0.95	1.00	0.90	0.68	0.96	1.00	0.97	0.97	0.97	1.00	0.96	0.98

(continued on next page)

Table 2 (continued)

Panel C: Correlation coefficients among different versions of pricing factors																	
		HML				RMW				CMA							
		Independent		Dependent		Independent		Dependent		Independent		Dependent					
		2 × 3	2 × 2	2 × 3	2 × 2	2 × 3	2 × 2	2 × 3	2 × 2	2 × 3	2 × 2	2 × 3	2 × 2				
Dependent	2 × 3	0.93	0.90	1.00	0.79	0.98	0.97	1.00	0.96	0.99	0.96	1.00	0.97				
	2 × 2	0.73	0.68	0.79	1.00	0.95	0.97	0.96	1.00	0.96	0.98	0.97	1.00				
		WML				IML				BMS							
		Independent		Dependent		Independent		Dependent		Independent		Dependent					
		2 × 3	2 × 2	2 × 3	2 × 2	2 × 3	2 × 2	2 × 3	2 × 2	2 × 3	2 × 2	2 × 3	2 × 2				
Independent	2 × 3	1.00	0.98	0.99	0.98	1.00	0.99	0.98	0.98	1.00	0.98	0.98	0.96				
	2 × 2	0.98	1.00	0.98	0.99	0.99	1.00	0.98	0.98	0.98	1.00	0.98	0.98				
Dependent	2 × 3	0.99	0.96	1.00	0.97	0.98	0.98	1.00	0.99	0.98	0.98	1.00	0.98				
	2 × 2	0.96	0.98	0.97	1.00	0.98	0.98	0.99	1.00	0.96	0.98	0.98	1.00				
Panel D: Correlation coefficients between different pricing factors																	
<i>Independent sort</i>																	
		2 × 3 factors							2 × 2 factors								
		$R_m - R_f$	SMB	HML	RMW	CMA	WML	IML	BMS	$R_m - R_f$	SMB	HML	RMW	CMA	WML	IML	BMS
$R_m - R_f$		1.00	-0.11	-0.35	-0.67	-0.05	-0.09	-0.63	0.55	1.00	-0.12	-0.33	-0.66	-0.11	-0.04	-0.62	0.53
SMB		-0.11	1.00	0.40	-0.15	-0.01	-0.05	-0.24	0.14	-0.12	1.00	0.43	-0.11	0.01	-0.08	-0.20	0.16
HML		-0.35	0.40	1.00	0.20	0.35	-0.18	0.22	0.03	-0.33	0.43	1.00	0.19	0.27	-0.14	0.12	-0.01
RMW		-0.67	-0.15	0.20	1.00	-0.18	0.29	0.74	-0.65	-0.66	-0.11	0.19	1.00	-0.12	0.23	0.67	-0.61
CMA		-0.05	-0.01	0.35	-0.18	1.00	-0.63	0.28	0.44	-0.11	0.01	0.27	-0.12	1.00	-0.60	0.40	0.38
WML		-0.09	-0.05	-0.18	0.29	-0.63	1.00	0.04	-0.71	-0.04	-0.08	-0.14	0.23	-0.60	1.00	-0.06	-0.64
IML		-0.63	-0.24	0.22	0.74	0.28	0.04	1.00	-0.49	-0.62	-0.20	0.12	0.67	0.40	-0.06	1.00	-0.45
BMS		0.55	0.14	0.03	-0.65	0.44	-0.71	-0.49	1.00	0.53	0.16	-0.01	-0.61	0.38	-0.64	-0.45	1.00
<i>Dependent sort</i>																	
		2 × 3 factors							2 × 2 factors								
		$R_m - R_f$	SMB	HML	RMW	CMA	WML	IML	BMS	$R_m - R_f$	SMB	HML	RMW	CMA	WML	IML	BMS
$R_m - R_f$		1.00	-0.09	-0.22	-0.68	-0.03	-0.10	-0.61	0.55	1.00	-0.12	-0.26	-0.65	-0.09	-0.06	-0.60	0.53
SMB		-0.09	1.00	0.37	-0.10	0.03	-0.09	-0.29	0.15	-0.12	1.00	0.09	-0.06	0.05	-0.10	-0.21	0.14
HML		-0.22	0.37	1.00	0.10	0.42	-0.20	0.15	0.11	-0.26	0.09	1.00	0.11	0.51	-0.19	0.38	0.03
RMW		-0.68	-0.10	0.10	1.00	-0.20	0.29	0.72	-0.65	-0.65	-0.06	0.11	1.00	-0.19	0.26	0.62	-0.62
CMA		-0.03	0.03	0.42	-0.20	1.00	-0.63	0.27	0.46	-0.09	0.05	0.51	-0.19	1.00	-0.63	0.39	0.42
WML		-0.10	-0.09	-0.20	0.29	-0.63	1.00	0.03	-0.71	-0.06	-0.10	-0.19	0.26	-0.63	1.00	-0.06	-0.66
IML		-0.61	-0.29	0.15	0.72	0.27	0.03	1.00	-0.46	-0.60	-0.21	0.38	0.62	0.39	-0.06	1.00	-0.40
BMS		0.55	0.15	0.11	-0.65	0.46	-0.71	-0.46	1.00	0.53	0.14	0.03	-0.62	0.42	-0.66	-0.40	1.00

stock sorting produces even larger differences in average momentum spreads, amounting to 0.32% ($t = 2.41$) and 0.47% ($t = 2.76$). The choice of independent or dependent sorting of portfolios has impact on differences in average profitability factor spreads (RMW) and default factor spreads (BMS). The average RMW_{S-B} are less than 1.19 standard errors from zero for the independent sorting, but more than 1.79 standard errors from zero for dependent sorted portfolios. The average BMS_{S-B} are 1.63 and 1.87 standard errors from zero for dependent sorted portfolios and less than 1.3 standard errors from zero for independent portfolio sorts. In contrast to the evidence of Fama and French (2016b), the results show small differences in investment premium CMA between small and big stocks. The largest CMA_{S-B} is -0.20% ($t = -1.62$ and non-significant) for dependent 2×2 stock portfolios.

We also show correlations between different versions of the same factor in Panel C of Table 2 and the correlations between all eight factors of the model for the all the various stock allocation and portfolio sorting approaches in Panel D of Table 2. The results show high correlations between different versions of the same factor, with the correlation coefficients ranging between 0.96 and 0.99. This leads us to believe that the impact of stock allocation or portfolio sorting approach on factor correlations seems to be low. The only exception seems to be the value pricing factor, since the correlations are lower. Namely, the correlation between HML factors calculated based on the independent 2×3 and 2×2 stock portfolios is 0.95, whereas the correlation between the same factors based on the dependent portfolio sorts amounts to 0.79. The correlation between independent and dependent 2×3 HML factors is 0.93 and 0.68 between independent and dependent 2×2 HML factors.

Finally, the results in Panel D of Table 2 show low impact of stock allocation choice or portfolio sorting approaches on calculated correlation coefficients among different factors of the eight-factor model. Furthermore, the factors of Fama and French (1993) three-factor model exhibit low correlation, since the highest correlation between SMB and HML factors amounts to 0.43. The first additional factor of the Fama and French (2016b) five-factor model RMW exhibits higher correlation with market factor in the range between 0.65 and 0.68, whereas the second investment factor CMA is the most correlated with value factor HML with correlation coefficients between 0.27 and 0.51. Among the additional three factors, that we add to the Fama and French five-factor model, momentum factor WML exhibits higher correlation with the investment factor

Table 3Regressions explaining the selected pricing factor with the other seven factors (based on 2×3 independent sorts) in the period January 1985–December 2016.

$R_m - R_f$	Inter.	$R_m - R_f$	SMB	HML	RMW	CMA	WML	IML	BMS	R^2
<i>b</i>	1.35		−0.53	−0.24	−0.58	−0.19	0.41	−0.15	0.91	0.63
<i>t(b)</i>	9.46		−5.84	−2.35	−5.45	−1.62	6.24	−1.79	9.17	
<i>SMB</i>										
<i>b</i>	0.25	−0.16		0.45	−0.15	−0.15	0.07	−0.13	0.15	0.33
<i>t(b)</i>	2.96	−5.84		8.86	−2.57	−2.33	1.74	−2.89	2.61	
<i>HML</i>										
<i>b</i>	0.25	−0.06	0.39		0.27	0.40	0.03	−0.09	0.04	0.39
<i>t(b)</i>	3.15	−2.35	8.86		4.95	7.21	0.83	−2.11	0.63	
<i>RMW</i>										
<i>b</i>	0.20	−0.13	−0.11	0.23		−0.52	−0.03	0.47	−0.04	0.76
<i>t(b)</i>	2.78	−5.45	−2.57	4.95		−10.96	−0.78	15.31	−0.85	
<i>CMA</i>										
<i>b</i>	0.15	−0.04	−0.10	0.30	−0.47		−0.15	0.44	0.22	0.68
<i>t(b)</i>	2.14	−1.62	−2.33	7.21	−10.96		−4.93	14.73	4.61	
<i>WML</i>										
<i>b</i>	0.02	0.23	0.12	0.06	−0.06	−0.41		−0.08	−1.05	0.68
<i>t(b)</i>	0.15	6.24	1.74	0.83	−0.78	−4.93		−1.32	−17.36	
<i>IML</i>										
<i>b</i>	0.08	−0.06	−0.16	−0.13	0.81	0.84	−0.06		−0.38	0.78
<i>t(b)</i>	0.78	−1.79	−2.89	−2.11	15.31	14.73	−1.32		−5.91	
<i>BMS</i>										
<i>b</i>	−0.23	0.20	0.11	0.03	−0.04	0.25	−0.42	−0.23		0.80
<i>t(b)</i>	−3.14	9.17	2.61	0.63	−0.85	4.61	−17.36	−5.91		

in the range between -0.60 and -0.63 . The liquidity factor *IML* is the most correlated with the market factor (between -0.60 and -0.62) and the profitability factor (0.62 – 0.74), while the default factor *BMS* is the most correlated with the momentum factor (from -0.64 to -0.71) and the profitability factor (from -0.61 to -0.65).

6. Multi-factor analysis of factor returns

We continue to further explore the interplay between explanatory factors by regressing each of the eight factors on the other seven factors. In this way we examine the average factor return, which is captured by its exposure to other explanatory factors. The regressions with explanatory factors constructed from different stock allocation and portfolio sorting approaches result in similar conclusions. Thus, we show only the regression results for the pricing factors based on the independently sorted 2×3 stock portfolios in Table 3.

The average return in regression of $R_m - R_f$ that is unexplained by other seven factors is 1.34% with t-statistics of 9.46. The intercepts are more than three standard deviations from zero in the regressions with *HML* and *BMS* factors as dependent variables and more than two standard deviations in the regressions with *SMB*, *RMW* and *CMA* factors as dependent variables. In the *WML* and *IML* regressions the intercepts are 0.02% ($t = 0.15$) and 0.08% ($t = 0.78$) respectively. The variation in *WML* factor is mostly absorbed by slopes of $R_m - R_f$, *CMA* and *BMS* factors. The positive market factor slope in the *WML* regression supports the prediction of positive correlation between momentum effect and market returns, whereas strongly negative *CMA* and *BMS* factor slopes indicate negative correlation with the momentum effect. In the regression with the *IML* factor as a dependent variable, the slope of *RMW* and *CMA* factors absorb most of the variation in *IML* factor return. The strongly positive *RMW* and *CMA* factor slopes indicate potentially strong relationship between stock liquidity, profitability and investment intensity.⁴

6.1. Performance evaluation of multi-factor models

We now turn our attention to the main aim of our study, testing the performance of extended versions of the Fama–French five factor model with the suggested additional pricing factors. Previously constructed explanatory factors, as described in previous sections, represent the main input for the performance analysis. However, investigating the performance for all the versions of the explanatory (RHS) factors that we can obtain due to various estimation choices in construction of testing (LHS) portfolios, different stock allocation breakpoints (number of portfolios) and portfolio sorting approaches, would make this a very tedious task with potentially little added value. Given our earlier findings regarding the impact of the pricing factors' construction choices, we thus restrict the analysis to six sets of 25 double sorted portfolios and we choose only explanatory factors based on the independent 2×3 portfolio sorts. We construct a range of three- to eight-factor models.

⁴ Alternatively, it could also indicate an omitted variable bias and/or endogeneity issues, but we are not making any conclusions here regarding the causality.

We begin the performance analysis by examining the general ability of each of the proposed multi-factor models to explain the average variation in portfolio returns. We monitor the impact of switching one of the explanatory factors with another, as well as adding additional explanatory factor in the model on the absolute value of intercepts. The decrease of the absolute value of an intercept implies better performance of the model by leaving less of the average variation in portfolio returns unexplained. To test whether the intercepts are significantly different from zero, we apply the well-known *GRS* test statistics of Gibbons et al. (1989).⁵

We continue the analysis of the performance of multi-factor models with a closer examination of each of the regression intercepts and factor slopes. Following the standard approach to asset pricing tests in earlier studies, we use Fama-MacBeth two step regression to estimate factor loadings, i.e. sensitivity coefficients, for each of the risk factors and factor risk premia. The first step of regression is executed on monthly excess returns (over the three month US Treasury bill rate) for portfolios based on 5×5 sorts. In fact, we run six sets of multi-factor regressions on 25 portfolios created from size variable and the second sorting variable (i.e. profitability, investment, momentum, liquidity and default risk). Based on the earlier findings that factors *CMA*, *WML* or *IML* (see Section 6) are mostly absorbed by their exposures to other explanatory factors, we include the orthogonalized versions of these factors in the regression analysis (*CMAO*, *WMLO* or *IMLO*). The orthogonalized version of the pricing factor is the sum of an intercept and residual in regressions of the original factor on the other explanatory factors as discussed in Section 6. Thus, the first step of Fama-MacBeth procedure envisages time-series regression with rolling window of 48 months and the explanatory factors *SMB*, *HML*, *RMW*, *CMAO*, *WMLO*, *IMLO* and *BMS* factors. The regression results provide the estimated factor coefficients for each month in the period from January 1989 to December 2016.

6.2. Regression results

6.2.1. Comparison of the overall performance

The results of the *GRS* test statistics and the absolute values of intercepts provide for the comparison of relative performance of multi-factor models. Table 4 provides the results for a list of multi-factor models subject to performance analysis: (i) Fama-French three-factor model with market, *SMB* and *HML* factors, (ii) five four-factor models as extensions of Fama-French three-factor model with the additional *RMW*, *CMA*, *WML*, *IML* or *BMS* factor, (iii) ten five-factor models as extensions of Fama-French three-factor models with a combination of the two of the *RMW*, *CMA*, *WML*, *IML* and *BMS* factors, (iv) ten six-factor models as extensions of the three-factor models with combinations of the three other factors, (v) five seven-factor models as extensions of three-factor models with a combination of the four additional factors, (vi) the proposed eight-factor model with the inclusion of all eight factors.

The absolute values of intercepts and *GRS* statistics of the proposed multi-factor models are calculated for all six sets of 25 portfolios created based on portfolio sorting on size and the second sorting variable. The incorporation of additional factors in the model mostly improves its performance, which is reflected in the lower *GRS* test value and the lower absolute value of the intercept. The absolute intercepts are between 0.63% and 0.70% for three-factor model and between 0.42% and 0.45% for eight-factor model. The *GRS* test values are between 7.75 and 9.06 for the three-factor model and between 4.75 and 6.20 for the eight-factor model. The *p*-values of the *GRS* test of the eight-factor model are still very low, suggesting the intercepts of the regressions are still statistically significantly different from zero. Although the extensions of the three- or five-factor models with additional factors improve their explanatory power, the proposed eight-factor model is still unable to capture all the variation in average portfolio returns.

The *GRS* test results and absolute intercepts depend on the selection of variables used to construct the testing portfolios. The results in Table 4 show that the eight-factor model performs the best for portfolios sorted on size and B/M (*GRS* = 4.75), size and investment (*GRS* = 4.82) or size and profitability (*GRS* = 5.09). The *GRS* test produces the highest values for portfolios sorted on size and liquidity (*GRS* = 6.20) or size and momentum (*GRS* = 5.67). The examination of absolute intercepts leads to the same conclusions regarding the performance of eight-factor models based on various portfolio sorts, although the differences in absolute intercepts between different sets of 25 portfolios are less pronounced.

Moreover, the results in Table 4 show the impact of explanatory factors selection on the performance of multi-factor models. Among the presented four-factor models, the extension of Fama and French (1993) three-factor model with *RMW* or *BMS* factor produces the lowest *GRS* test values and the lowest absolute values of alphas in all six sets of 25 portfolios. The reductions of the absolute value of the alpha after the incorporation of additional factor in the model are between 14 and 17 basis points. The extension of three factor model with both, *RMW* and *BMS* factor, results in the best model performance of all the ten five-factor models in our study. This five-factor model improves the absolute alpha of the three-factor model by 19 to 23 basis points, whereas the improvement of the alpha of the best performing four-factor model amounts to approximately 5 basis points. Measured in terms of both criteria, the *GRS* test and the absolute value of the alpha, the extensions of Fama and French three-factor model with additional two factors indicate that five of five-factor models seem to provide better performance than Fama and French five-factor model. Among the proposed six factor models, the extension of Fama and French five-factor model with *BMS* factor exhibits the best performance. This model further reduces the absolute

⁵ The test has been proposed by Gibbons et al. (1989). It relates to the joint hypothesis that intercepts or α s in time series Fama-MacBeth regressions are equal to 0. A fully specified multi-factor model would yield an intercept statistically non-significantly different from 0. One can argue that the reduction in the value of the intercept (hence, the value of *GRS* test statistics) implies and improvement of the multi-factor model, as a smaller part of the returns is left "unexplained" by the proposed factors.

Table 4Performance parameters of various multi-factor models (based on 2×3 independent sorts) in the period January 1995 – December 2016.

	Size-B/M		Size-Profitability		Size-Investment		Size-Momentum		Size-Liquidity		Size-Default risk	
	GRS	$ \alpha_i $	GRS	$ \alpha_i $	GRS	$ \alpha_i $	GRS	$ \alpha_i $	GRS	$ \alpha_i $	GRS	$ \alpha_i $
<i>Three factor model</i>												
FF three-factor model (Rm-Rf, SMB, HML)	7.75	0.69	7.95	0.63	8.07	0.63	9.06	0.66	8.41	0.70	8.28	0.66
<i>Four-factor models</i>												
Rm-Rf, SMB, HML, RMW	5.94	0.54	6.20	0.51	5.94	0.51	6.80	0.54	6.18	0.55	6.25	0.53
Rm-Rf, SMB, HML, CMA	7.79	0.68	8.50	0.63	8.11	0.63	9.18	0.65	8.58	0.69	8.52	0.65
Rm-Rf, SMB, HML, WML	7.00	0.64	7.00	0.60	7.18	0.60	8.10	0.61	7.38	0.65	7.49	0.61
Rm-Rf, SMB, HML, IML	6.25	0.62	6.34	0.58	6.42	0.59	7.51	0.63	7.71	0.64	7.23	0.61
Rm-Rf, SMB, HML, BMS	6.00	0.51	5.83	0.48	5.99	0.49	6.81	0.49	6.17	0.52	6.64	0.49
<i>Five-factor models</i>												
Rm-Rf, SMB, HML, RMW, CMA	5.82	0.54	6.27	0.51	5.91	0.51	6.84	0.54	6.42	0.56	6.58	0.53
Rm-Rf, SMB, HML, RMW, WML	5.80	0.53	5.99	0.50	5.78	0.51	6.59	0.51	5.93	0.54	6.18	0.52
Rm-Rf, SMB, HML, RMW, IML	5.58	0.54	5.79	0.51	5.59	0.52	6.66	0.55	6.95	0.55	6.37	0.52
Rm-Rf, SMB, HML, RMW, BMS	5.41	0.47	5.46	0.45	5.27	0.45	6.05	0.45	5.44	0.47	6.00	0.45
Rm-Rf, SMB, HML, CMA, WML	6.77	0.64	7.46	0.61	7.15	0.60	7.97	0.61	7.55	0.66	7.39	0.61
Rm-Rf, SMB, HML, CMA, IML	5.95	0.59	6.38	0.56	6.10	0.56	7.20	0.59	7.34	0.60	6.88	0.58
Rm-Rf, SMB, HML, CMA, BMS	5.53	0.49	5.90	0.48	5.73	0.47	6.54	0.47	6.00	0.50	6.18	0.47
Rm-Rf, SMB, HML, WML, IML	5.54	0.57	5.49	0.55	5.63	0.55	6.60	0.55	7.10	0.59	6.33	0.56
Rm-Rf, SMB, HML, WML, BMS	5.88	0.49	5.71	0.47	5.88	0.46	6.77	0.47	6.09	0.49	6.52	0.47
Rm-Rf, SMB, HML, IML, BMS	5.12	0.48	5.02	0.46	5.09	0.47	6.06	0.48	6.46	0.49	5.90	0.47
<i>Six-factor models</i>												
Rm-Rf, SMB, HML, RMW, CMA, WML	5.29	0.52	5.79	0.50	5.45	0.50	6.18	0.51	5.97	0.54	5.94	0.51
Rm-Rf, SMB, HML, RMW, CMA, IML	5.57	0.54	5.94	0.51	5.63	0.52	6.70	0.55	6.93	0.55	6.44	0.53
Rm-Rf, SMB, HML, RMW, CMA, BMS	4.79	0.43	5.05	0.41	4.83	0.42	5.59	0.43	5.24	0.44	5.45	0.42
Rm-Rf, SMB, HML, RMW, WML, IML	5.28	0.53	5.43	0.51	5.28	0.51	6.26	0.52	6.77	0.54	5.97	0.52
Rm-Rf, SMB, HML, RMW, WML, BMS	5.32	0.45	5.39	0.43	5.20	0.43	6.02	0.44	5.37	0.45	5.91	0.43
Rm-Rf, SMB, HML, RMW, IML, BMS	5.00	0.46	5.04	0.45	4.89	0.45	5.88	0.46	6.30	0.47	5.71	0.45
Rm-Rf, SMB, HML, CMA, WML, IML	5.53	0.57	6.00	0.55	5.76	0.55	6.68	0.55	7.12	0.59	6.36	0.56
Rm-Rf, SMB, HML, CMA, WML, BMS	5.51	0.48	5.98	0.47	5.75	0.46	6.53	0.46	6.08	0.50	6.16	0.46
Rm-Rf, SMB, HML, CMA, IML, BMS	5.06	0.48	5.34	0.46	5.18	0.46	6.11	0.47	6.40	0.49	5.85	0.47
Rm-Rf, SMB, HML, WML, IML, BMS	5.10	0.48	5.01	0.46	5.10	0.47	6.05	0.47	6.54	0.49	5.89	0.47
<i>Seven-factor models</i>												
Rm-Rf, SMB, HML, RMW, CMA, WML, IML	5.16	0.52	5.61	0.50	5.30	0.51	6.19	0.51	6.73	0.54	5.93	0.51
Rm-Rf, SMB, HML, RMW, CMA, WML, BMS	4.78	0.43	5.14	0.41	4.85	0.42	5.59	0.43	5.33	0.44	5.44	0.42
Rm-Rf, SMB, HML, RMW, CMA, IML, BMS	4.76	0.43	5.01	0.42	4.79	0.42	5.69	0.43	6.09	0.45	5.49	0.43
Rm-Rf, SMB, HML, RMW, WML, IML, BMS	4.98	0.46	5.05	0.44	4.89	0.44	5.86	0.45	6.36	0.47	5.69	0.45
Rm-Rf, SMB, HML, CMA, WML, IML, BMS	5.05	0.48	5.42	0.46	5.21	0.47	6.09	0.47	6.52	0.49	5.85	0.47
<i>Eight-factor model</i>												
Rm-Rf, SMB, HML, RMW, CMA, WML, IML, BMS	4.75	0.44	5.09	0.42	4.82	0.43	5.67	0.44	6.20	0.45	5.49	0.49

value of alpha of the best performing five-factor model (market, *SMB*, *HML*, *RMW* and *BMS* factors) by 2 to 4 basis points. The improvements in the GRS test values are between 0.41 and 0.62. Other six-factor models exhibiting high performance incorporate *IML* or *WML* factor next to the market, *SMB*, *HML*, *RMW* and *BMS* factors. The further extension of the six-factor model with additional factor has a smaller positive impact on the model performance. Seven-factor models, which omit *IML* or *WML* factor of the proposed eight factors in our study, yield the lowest absolute value of alpha and the GRS test statistics. The proposed eight-factor model shows similar model performance according to the GRS test and the absolute value of alpha to the that of the best performing seven-factor models.

6.2.2. Regression intercepts and factor slopes

The examination of regression intercepts and factor slopes provides further insight into the relative performance of various multi-factor models. Table 5 shows the average intercepts and factor slopes with t-statistics for the three-factor model with market, *SMB* and *HML* factors, the five-factor model as extension of the three-factor model with *RMW* and *CMAO* factors and the proposed eight-factor model with orthogonal factors *CMAO*, *WML* and *IMLO*. The tests of multi-factor model performance for all six sets of 5×5 sorted portfolios lead to similar conclusions. Thus, we restrict the presentation and the discussion of regression results only to 25 Size-B/M portfolios.

Table 5 shows that extending the Fama-French three-factor model with *RMW* and *CMAO* factors and further with *WML*, *IMLO* and *BMS* factors reduces the average regression intercepts on 25 size-B/M portfolios. The three-factor model yields regression intercept values in the range between -0.31% and -1.00% (Panel A of Table 5), the five-factor model between -0.15% and -0.83% (Panel B of Table 5) and the eight-factor model between -0.02% and -0.61% (Panel C of Table 5). The t-statistics of regression intercepts of the eight-factor model are between -0.03 and -1.76 , which implies that the proposed eight-factor model captures most of the variation in average returns of portfolios sorted on size and value. Moreover,

Table 5

Time-series regressions for 25 Size-B/M portfolios; January 1985 – December 2016; Table provides intercepts for three- and five-factor model and intercepts and coefficients for eight-factor model.

	Low	2	3	4	High	Low	2	3	4	High
Panel A: 3-factor model: $R_{it} - R_{ft} = a + b(R_{mt} - R_{ft}) + sSMB_t + hHML_t$										
	a					$t(a)$				
Small	-0.94	-0.55	-0.54	-0.51	-0.37	-1.45	-0.90	-1.27	-1.64	-1.30
2	-0.91	-1.00	-0.76	-0.66	-0.69	-1.91	-2.60	-2.30	-2.06	-1.89
3	-0.88	-0.58	-0.57	-0.57	-0.67	-2.08	-1.59	-1.63	-1.63	-1.64
4	-0.58	-0.51	-0.47	-0.47	-0.50	-1.51	-1.79	-1.53	-1.46	-1.17
Big	-0.31	-0.59	-0.48	-0.68	-0.70	-1.64	-2.62	-1.81	-2.01	-1.57
Panel B: 5-factor model: $R_{it} - R_{ft} = a + b(R_{mt} - R_{ft}) + sSMB_t + hHML_t + rRMW_t + cCMAO_t$										
	a					$t(a)$				
Small	-0.69	-0.39	-0.41	-0.46	-0.15	-1.02	-0.61	-0.93	-1.42	-0.55
2	-0.74	-0.83	-0.68	-0.64	-0.49	-1.48	-2.06	-1.96	-1.84	-1.29
3	-0.65	-0.39	-0.39	-0.45	-0.43	-1.46	-1.02	-1.07	-1.22	-1.01
4	-0.28	-0.45	-0.38	-0.33	-0.23	-0.74	-1.50	-1.18	-0.99	-0.52
Big	-0.27	-0.48	-0.37	-0.55	-0.65	-1.36	-2.03	-1.33	-1.51	-1.38
Panel C: 8-factor model: $R_{it} - R_{ft} = a + b(R_{mt} - R_{ft}) + sSMB_t + hHML_t + rRMW_t + cCMAO_t + wWMLO_t + iIMLO_t + dBMS_t$										
	a					$t(a)$				
Small	-0.20	-0.09	-0.20	-0.29	-0.03	-0.29	-0.13	-0.43	-0.84	-0.12
2	-0.54	-0.61	-0.61	-0.60	-0.36	-1.10	-1.48	-1.75	-1.66	-0.93
3	-0.39	-0.30	-0.25	-0.39	-0.24	-0.90	-0.79	-0.67	-0.99	-0.54
4	-0.14	-0.35	-0.29	-0.24	-0.02	-0.36	-1.13	-0.93	-0.70	-0.03
Big	-0.20	-0.41	-0.32	-0.50	-0.53	-0.94	-1.76	-1.18	-1.43	-1.13
	s					$t(s)$				
Small	2.11	1.72	1.49	1.15	1.23	4.72	3.89	5.07	5.06	6.38
2	1.79	1.51	1.43	1.31	1.36	5.54	5.57	6.34	5.56	5.39
3	1.12	1.08	0.79	0.68	0.71	3.90	4.40	3.33	2.74	2.55
4	0.36	0.20	0.19	0.18	0.09	1.52	1.02	0.95	0.82	0.31
Big	-0.13	-0.04	-0.01	-0.15	-0.32	-0.95	-0.30	-0.03	-0.67	-1.06
	h					$t(h)$				
Small	-1.09	-0.22	-0.07	0.37	0.78	-2.34	-0.48	-0.24	1.56	3.83
2	-0.74	-0.43	0.04	0.24	0.54	-2.21	-1.53	0.17	0.97	2.06
3	-0.68	-0.44	0.07	0.25	0.58	-2.30	-1.71	0.30	0.97	1.98
4	-0.52	-0.10	-0.02	0.34	0.53	-2.08	-0.49	-0.09	1.47	1.73
Big	-0.27	-0.01	0.11	0.33	0.81	-1.86	-0.08	0.61	1.45	2.58
	r					$t(r)$				
Small	-0.30	-0.25	0.06	-0.03	-0.36	-0.70	-0.59	0.19	-0.12	-1.97
2	-0.47	-0.19	-0.12	-0.03	-0.29	-1.51	-0.73	-0.56	-0.13	-1.23
3	-0.30	-0.32	-0.17	-0.09	-0.26	-1.11	-1.38	-0.75	-0.39	-0.98
4	-0.36	-0.17	-0.12	-0.17	-0.19	-1.56	-0.92	-0.63	-0.79	-0.67
Big	-0.09	-0.20	-0.07	-0.05	-0.03	-0.70	-1.36	-0.44	-0.23	-0.09
	c					$t(c)$				
Small	-0.81	-1.03	-0.44	-0.47	-0.38	-0.99	-1.30	-0.84	-1.16	-1.08
2	-1.04	-1.23	-0.61	-0.17	-0.29	-1.78	-2.50	-1.50	-0.39	-0.62
3	-1.28	-0.90	-0.66	-0.30	-0.15	-2.48	-1.98	-1.52	-0.66	-0.28
4	-1.36	-0.32	-0.22	-0.02	-0.13	-3.10	-0.89	-0.60	-0.06	-0.24
Big	-0.42	-0.07	0.04	0.43	0.78	-1.69	-0.25	0.14	1.07	1.44
	w					$t(w)$				
Small	0.02	0.33	-0.05	0.13	0.11	0.06	0.95	-0.20	0.73	0.69
2	0.34	0.31	0.35	0.22	0.26	1.31	1.43	1.88	1.17	1.27
3	0.44	0.43	0.37	0.28	0.23	1.92	2.12	1.89	1.36	1.00
4	0.42	0.29	0.40	0.26	0.26	2.09	1.76	2.37	1.41	1.09
Big	0.07	0.15	0.13	0.21	0.05	0.62	1.23	0.94	1.16	0.20
	i					$t(i)$				
Small	-0.81	-0.66	-0.09	-0.22	-0.15	-1.34	-1.12	-0.22	-0.72	-0.57
2	-1.04	-0.64	-0.22	-0.08	-0.11	-2.37	-1.74	-0.73	-0.23	-0.32
3	-0.73	-0.45	-0.32	-0.10	-0.11	-1.90	-1.34	-0.97	-0.30	-0.28
4	-0.67	-0.06	0.02	0.08	-0.03	-2.06	-0.22	0.07	0.26	-0.07
Big	-0.24	0.04	0.12	0.43	0.46	-1.31	0.23	0.54	1.46	1.15
	d					$t(d)$				
Small	0.68	0.26	0.29	0.20	0.18	1.75	0.64	1.11	0.98	1.08
2	-0.21	-0.05	-0.15	-0.05	0.08	-0.72	-0.22	-0.77	-0.23	0.37
3	-0.11	-0.20	-0.02	-0.03	0.20	-0.45	-0.95	-0.09	-0.12	0.82
4	-0.21	-0.07	0.10	0.13	0.45	-1.01	0.39	0.55	0.67	1.72
Big	-0.05	0.05	0.16	0.27	0.44	-0.41	0.41	1.03	1.44	1.71

the statistical significance of regression intercepts across 5×5 size and value portfolios implies that the unexplained part of return is most likely not associated with specific characteristics of the particular portfolio (i.e. small or big, value or growth).

The results in Panel C of Table 5 show the eight-factor model slopes for *SMB*, *HML*, *RMW*, *CMAO*, *WMLO*, *IMLO* and *BMS* factors. We do not show market factor slopes, which are close to 1.00 for all portfolios. In line with our earlier evidence based on *GRS* test analysis, the factor slopes confirm the impact of the approach to construction of testing (LHS) portfolios (portfolio sort variables) on regression results. The *SMB* factor slopes exhibit large positive factor loadings on small stock portfolio (from 1.15 to 2.11) and small negative loadings on big stock portfolios (from -0.32 to -0.04). The *HML* factor slopes are large and negative for growth stock portfolios, but positive for value stock portfolios. The results for remaining five sets of 5×5 portfolios (not presented in this paper for the sake of brevity) lead to similar conclusions. Namely, that the factor slopes correspond to the pertinent characteristics (sorting variables) used to create testing (LHS) portfolios. In addition, the results for factor slopes show less evident relationship when it comes to other pricing factors. The *CMAO* and *IMLO* factor slopes are negative, except for the big stock portfolios. Moreover, regression slopes for both factors are large and negative in low B/M portfolios, which indicates that growth portfolios stocks correspond to aggressive investment policies and positive premium for higher liquidity. The *RMW* pricing factor slopes are negative, while the *WMLO* are positive. However, both factor slopes show low t-statistics (mostly statistically not different from zero). The *BMS* factor slopes are negative for growth stocks and positive for value stocks, but, similar to profitability and momentum factors, mostly statistically not different from zero.

7. Conclusion

The proposed multi-factor model combines eight pricing factors, which, based on the past empirical evidence, are expected to capture most of the variation in average cross-sectional returns. The *GRS* test rejects the eight-factor model, which indicates that the model is unable to explain all the variation in returns on double sorted portfolios of stocks. However, the *GRS* test values and the values of the regression intercepts show better performance of the eight-factor model in comparison to the three-factor or five-factor model of Fama and French (1993, 2016b). Similarly, most of the six-factor and the seven-factor models created from the selection of factors of the eight-factor model exhibit better performance than models with five or less factors. Thus, momentum, liquidity and default risk pricing factors seem to significantly contribute to the explanation of average portfolio returns. However, the average factor slopes show that market, size and value factors are the most important in explaining the cross-section of average returns, whereas the contribution of other factors is less pronounced. Despite some specific characteristics of small stock and growth portfolios, the model performs considerably well across all 5×5 portfolios.

Our evidence suggests that the estimation choices in portfolio construction have some, albeit very limited, impact on explanatory factors and the performance of multi-factor models. We implement two stock allocation approaches based on different number of sorting portfolios – 2×3 and 2×2 portfolios. The first approach produces more pronounced factor returns by focusing more on the extreme returns in stock portfolios, but it results in the lower portfolio diversification due to the exclusion of the middle 40% stocks. The choice between independent and dependent portfolio sorting has the highest impact on value, profitability and momentum factors, but the end results are qualitatively very similar, in particular from the perspective of statistical significance. Moreover, the regression results support the prior evidence that the approach to creating the testing (LHS) portfolios affects the model performance and regression slopes. However, the tests of multi-factor models produce small differences in the *GRS* test values and the absolute value of alphas. We are left with mixed conclusions as to which approach to testing (LHS) portfolios construction assures the best multi-factor model performance. The regression results show that the variables used to form testing (LHS) portfolios (portfolio sorting variables) have the highest impact on slopes of explanatory factors, which are created from the same or correlated variables.

Our study shows that a strong relationship between stock (corporate) characteristics and return patterns or high factor returns does not necessarily result in a high explanatory power of the pricing factor in the multi-factor model. Finally, given that the eight-factor model still leaves quite some variation in average returns unexplained represents the motivation for the search of further improvements in the multi-factor model performance, by either redefining of some pricing factors or by continued search for additional pricing factors.

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